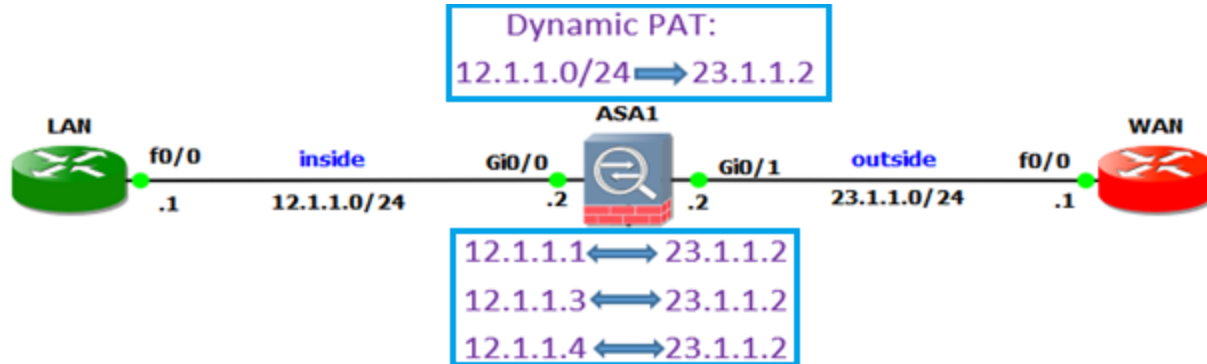


NAT on Cisco ASA 9.x:

- o NAT is a Cisco term, which stands for Network Address Translation.
- o NAT is a process that involves translating Private IP into Public IP addresses.
- o The process of translating one IP address to another IP is known as a NAT.
- o Network Address Translation (NAT) mostly happens on Cisco ASA firewall.
- o Router and Firewall is a device, which is used for network Address Translation.
- o NAT Configuration on ASA is completely different from configuration on Router.
- o Different versions of ASA have different NAT configurations, old v8.4 & new v9.x.
- o In ASA, there are many forms and kinds of Network Address Translation (NAT).
- o Network Address Translation is used to reduce the requirement of Public IP addresses.
- o Network Address Translation increases the security of Internal Computer Networks.
- o NAT increases security by hiding the internal network topology and addressing.
- o NAT translates Private IP into Public IP address & Public IP into Private IP address.
- o NAT is used to connect a device with Private IP address to the Internet or WAN.
- o Network Address Translation hides an organization's internal network from external.
- o Network Address Translation (NAT) modifies only the Layer 3 header of IP address.
- o PAT, translation of an IP address and Port to another IP address and Port number.
- o Port Address Translation (PAT) modifies both the Layer 3 and Layer 4 header of IP.



NAT:

Network Address Translation, or NAT, implies a translation of an IP address to another IP address. NAT modifies only the Layer 3 header of OSI reference Model.

PAT:

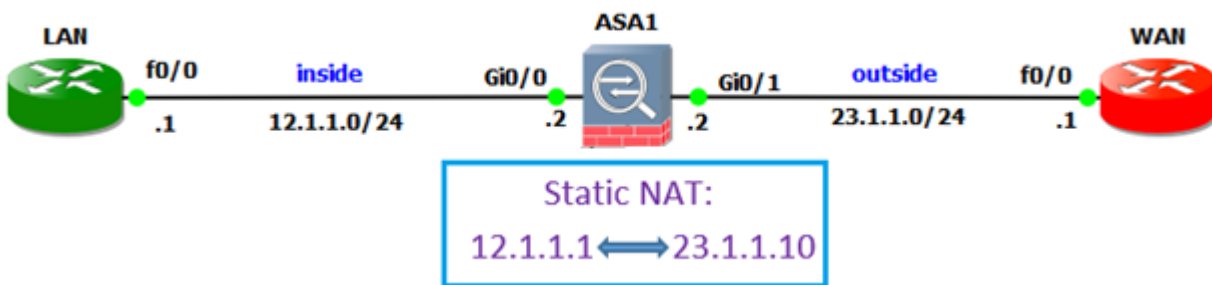
Port Address Translation, or PAT, implies a translation of an IP address and Port to another IP address and Port. PAT modifies both the Layer 3 and Layer 4 header. PAT as a subset of NAT.

Static vs Dynamic:

Both a NAT and a PAT can exist in two forms: Static NAT or Dynamic NAT.

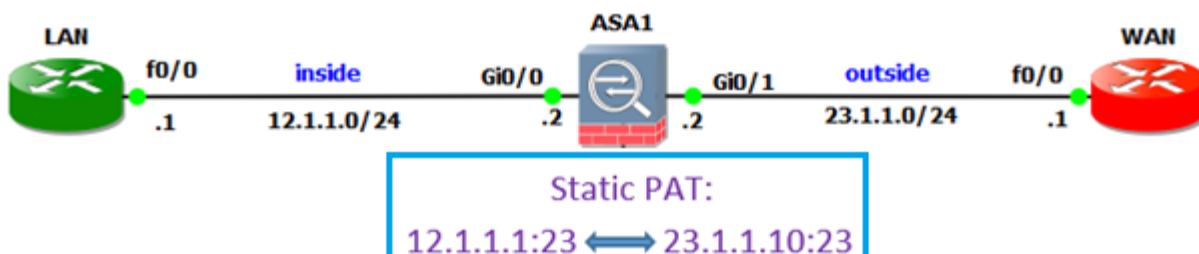
Static NAT:

- o A Static NAT mapping is sometimes referred to as a One-to-One translation.
- o Static NAT implies a translation of single IP address to another single IP address.
- o Static Network Address Translation modifies only the Layer 3 header of OSI Model.
- o Static NAT maps one address to another address server is located behind firewall.
- o In static NAT we create one to one mapping of IP addresses which is Bi-directional.
- o Mapping between IP on a more secure interface & an IP on a less secure interface.
- o Static NAT allows hosts on less secure interface to access hosts on higher security.
- o Without exposing the actual IP address of the host on the higher security interface.
- o The Static Network Address Translation (NAT), translation is the bidirectional NAT.
- o Whether internal host or external host sent first packet, it pass through Static NAT.
- o SNAT is used to do one-to-one mapping between inside address & outside address.
- o Static Network Address Translation allows connections from outside host to host.



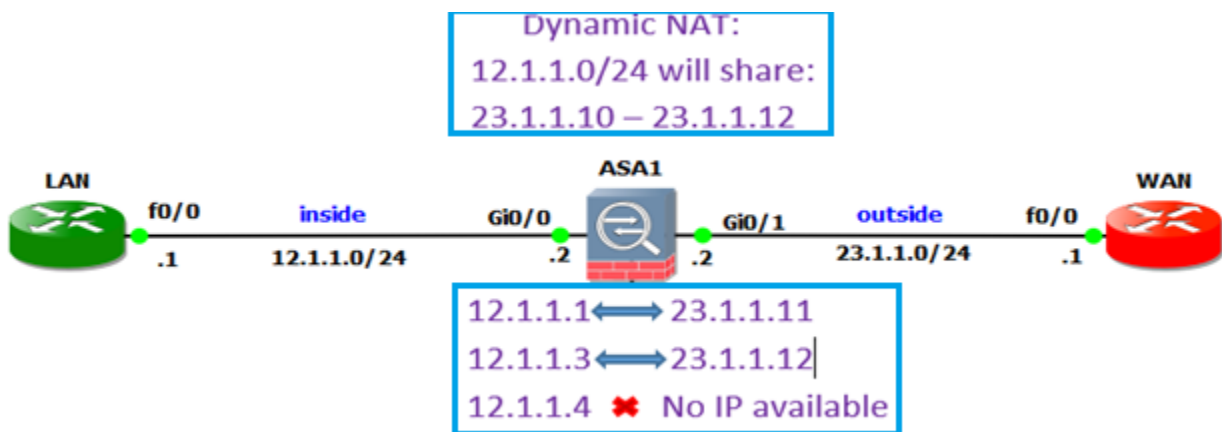
Static PAT:

- o Static PAT implies a translation of single IP address & Port to another single IP & Port.
- o Static PAT modifies both the Layer 3 and Layer 4 header of OSI Reference model.
- o It allow specific UDP or TCP port on global address to be translated to specific port.
- o In static PAT map port of one IP address with another IP address port Uni-directional.



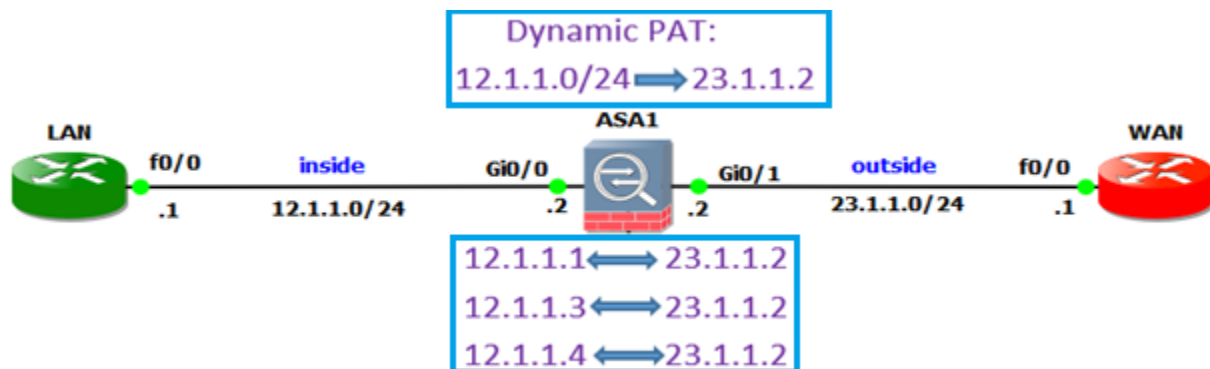
Dynamic NAT:

- o A Dynamic mapping is sometimes referred to as a One-to-Many translation.
- o Implying that in a Dynamic translation, many addresses can appear as one.
- o Dynamic NAT is used when you have a “pool” of public IP addresses to use.
- o Used when pool of public IP addresses want to assign internal hosts dynamically.
- o Translation of just IP address, where device determines new IP after translation.
- o In Dynamic Network Address Translation, we map multiple IP addresses to some.
- o A Dynamic NAT is a translation in which only the IP addresses are being modified.
- o Allows multiple internal hosts with Private IP to temporarily own dedicated Public IP.
- o This temporarily own dedicated public IP address as long as have an active session.



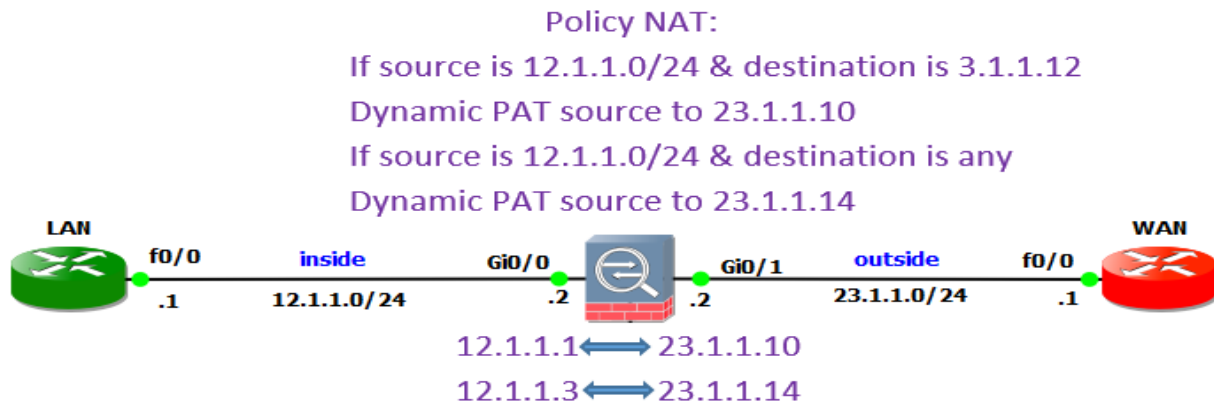
Dynamic PAT:

- o Translation of IP & Port, where device determines new IP & Port after translation.
- o Dynamic PAT is translation in which IP addresses & Port numbers are being modified.
- o Mapping between pre-translation & post-translation attributes is dynamically determined.
- o It allows multiple internal hosts with Private IP addresses to share one or more Public IP.



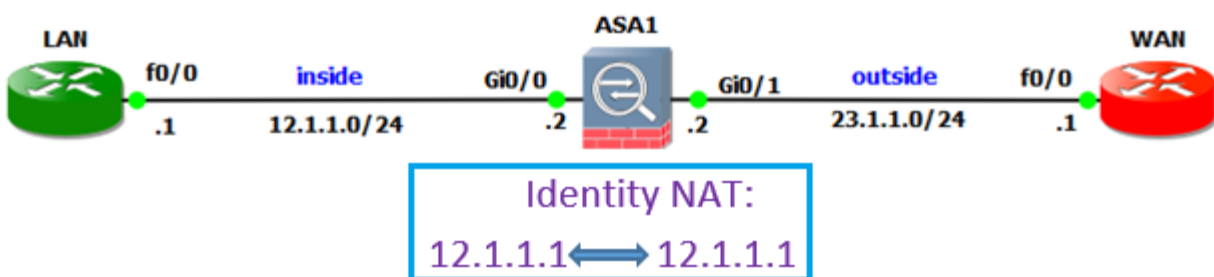
Policy NAT:

- o Sometimes might be beneficial to conditionally translate traffic based upon its destination.
- o Any NAT, That the translation decision is based upon both the Source and the Destination.
- o The configuration tells the device to translate packet based upon the criteria of matching.
- o Criteria matching is both the Source and Destination of the packet is called Policy NAT.
- o Policy NAT is any translation that occurs based upon matching both Source & Destination.
- o It is cannot be configured using Auto NAT syntax because Auto NAT only considers Source.



Identity NAT:

- o Traffic from inside interface to flow through to outside interface without changing address.
- o To translate a Real IP address to itself, essentially bypassing Network Address Translation.
- o Identity NAT is useful in VPN configuration to exempt VPN traffic from the NAT operation.
- o Configuration of Identity NAT involves re-using an object as both real object & mapped.



Real IP:

The word real indicates what is really configured on a System. Referring to the private IP that is assigned to the System.

Real Port:

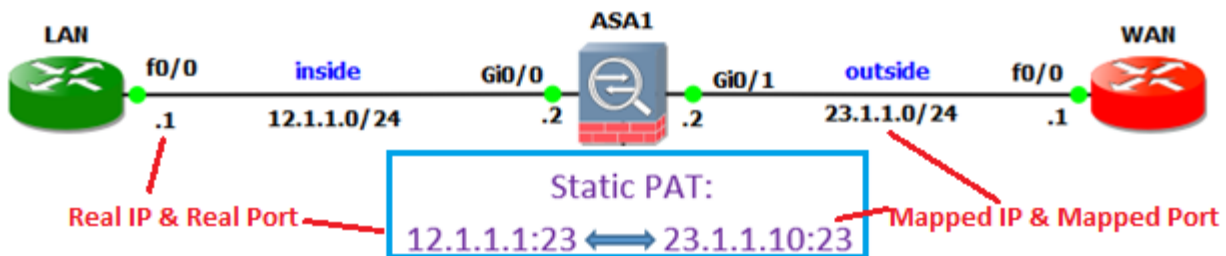
Referring to the port of service that actually running on the System.

Mapped IP:

Referring to the public IP that is accessible from Internet. The word mapped indicates attributes after a translation has occurred.

Mapped Port:

Referring to the port that is used for public access. Prefer to use real port the same as mapped port. The word mapped indicates attributes after a translation has occurred.



NAT Order:

NAT is order within three sections.

Section 1 – Manual NAT, Section 2 – Auto NAT and Section 3 – Manual NAT After-Auto.

By default, only sections 1 and 2 are used. However, if you need to place a manual NAT rule after Auto NAT you can specify the keyword **after-auto** when configuring a Manual NAT rule to place it within Section 3.

Manual NAT (Section-1):

Manual NAT can do everything that Auto NAT can, and do Policy NAT and Twice NAT. Manual NAT can make a NAT decision based upon the Source, or upon both the Source and Destination. Manual NAT can translate the Source, the Destination, or even both the Source and Destination at the same time.

Auto NAT (Section-2):

Auto NAT is configured within the Object definition. Auto NAT can be used anytime you need to make a NAT decision based upon only the Source of traffic. Which means each of the four types of translations (**Static NAT, Static PAT, Dynamic PAT, and Dynamic NAT**) can be configured with Auto NAT.

Manual NAT After-Auto (Section-3):

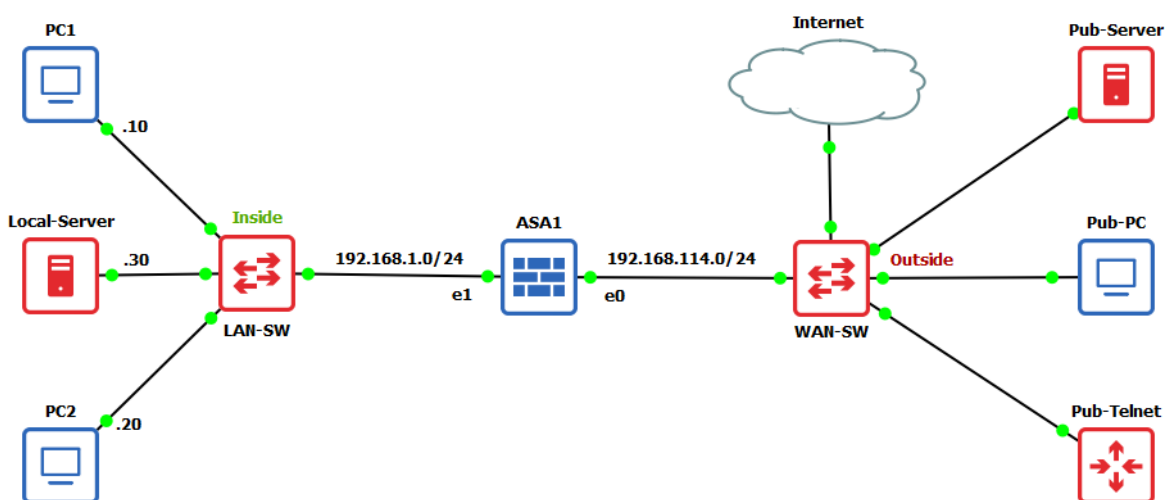
If to place, a manual NAT rule after Auto NAT specify the keyword after-auto when configuring a Manual NAT rule to place it within Section 3. It will be less preferred over an Auto NAT.

Objects:

An object represents any single item in the network environment. Two types of objects can be configured. A **network object**— represents one IP address, or one IP Subnet, or one IP address range. A **service object**— represents one set of a Protocol, Source Port, and/or Destination port.

object network WEB-SERVER host 192.168.1.30	object network INSIDE-NETWORK subnet 192.168.1.0 255.255.255.0
object network PUBLIC-IPs range 192.168.114.100 192.168.114.110	object service PROTOCOL service esp
object service PROT-DST service tcp destination eq 80	object service PROT-SRC service tcp source eq 23
object service PROT-SRC-DST service udp source eq 53 destination eq 53	

Lab Time:



Device Name	Configuration
PC1 Webterm Configuration	# Static config for eth0 auto eth0 iface eth0 inet static address 192.168.1.10 netmask 255.255.255.0 gateway 192.168.1.100 up echo nameserver 8.8.8.8 > /etc/resolv.conf

PC2 Webterm Configuration	# Static config for eth0 auto eth0 iface eth0 inet static address 192.168.1.20 netmask 255.255.255.0 gateway 192.168.1.100 up echo nameserver 8.8.8.8 > /etc/resolv.conf
Local-Server Toolbox Configuration	# Static config for eth0 auto eth0 iface eth0 inet static address 192.168.1.30 netmask 255.255.255.0 gateway 192.168.1.100 up echo nameserver 8.8.8.8 > /etc/resolv.conf
Pub-Server Toolbox Configuration	# DHCP config for eth0 auto eth0 iface eth0 inet dhcp
Pub-PC Webterm Configuration	# DHCP config for eth0 auto eth0 iface eth0 inet dhcp
Pub-Telnet Configuration	Pub-Telnet(config)#interface Ethernet0/0 Pub-Telnet(config-if)# ip address dhcp Pub-Telnet(config-if)#no shutdown Pub-Telnet(config-if)#exit Pub-Telnet(config)#line vty 0 4 Pub-Telnet(config-line)#transport input all Pub-Telnet(config-line)#password 123 Pub-Telnet(config-line)#login Pub-Telnet(config)#enable password 123

Cisco ASA Firewall Configuration
ASA1(config)# interface Ethernet0 ASA1(config-if)#nameif outside ASA1(config-if)#security-level 0 ASA1(config-if)#ip address 192.168.114.100 ASA1(config-if)#no shutdown
ASA1(config)# interface Ethernet1 ASA1(config-if)#nameif inside ASA1(config-if)#security-level 100 ASA1(config-if)#ip address 192.168.1.100 ASA1(config-if)#no shutdown
ASA1(config)# access-list 1 extended permit ip any any ASA1(config)# access-group 1 in interface outside

Static NAT with Auto NAT	
ASA1(config)# object network LAN	
ASA1(config-network-object)# host 192.168.1.10	
ASA1(config-network-object)#nat (inside,outside) static 192.168.114.101	
ASA1# show nat	ASA1# show run object
ASA1# show run nat	ASA1# show xlate
ASA1# show nat detail	root@PC1:~# ping 8.8.8.8
root@PC1:~# telnet Pub-Telnet	root@PC1:~# visit any website

Static NAT with Manual NAT	
ASA1(config)# object network LAN-Private	
ASA1(config-network-object)# host 192.168.1.10	
ASA1(config)# object network LAN-Public	
ASA1(config-network-object)# host 192.168.114.101	
ASA1(config)#nat (inside,outside) source static LAN-Private LAN-Public	
ASA1# show nat	ASA1# show run object
ASA1# show run nat	ASA1# show xlate
ASA1# show nat detail	root@PC1:~# ping 8.8.8.8
root@PC1:~# telnet Pub-Telnet	root@PC1:~# visit any website

Static PAT with Auto NAT	
ASA1(config)# clear configure nat	ASA1(config)# clear configure object
ASA1(config)# object network LAN-Web	
ASA1(config-network-object)# host 192.168.1.30	
ASA1(config-network-object)#nat (inside,outside) static 192.168.114.101 service tcp 80 80	
ASA1(config)# access-list 100 extended permit ip any any	
ASA1(config)# access-group 100 in interface outside	
ASA1# show nat	ASA1# show conn
ASA1# show run nat	ASA1# show xlate
ASA1# show nat detail	root@Pub-PC:~# http://192.168.114.101

Static PAT with Manual NAT	
ASA1(config)# clear configure nat	ASA1(config)# clear configure object
ASA1(config)# object network LAN-Private	
ASA1(config-network-object)# host 192.168.1.30	
ASA1(config)# object network LAN-Public	
ASA1(config-network-object)# host 192.168.114.101	
ASA1(config)# object service TCP80	
ASA1(config-service-object)# service tcp source eq 80	
ASA1(config)# nat (inside,outside) source static LAN-Private LAN-Public service TCP80 TCP80	
ASA1# show nat	ASA1# show conn
root@Pub-PC:~# http://192.168.114.101	ASA1# show xlate

Dynamic NAT with Auto NAT	
ASA1(config)# clear configure nat	ASA1(config)# clear configure object
ASA1(config)# object network WAN ASA1(config-network-object)# range 192.168.114.101 192.168.114.105	
ASA1(config)# object network LAN ASA1(config-network-object)# subnet 192.168.1.0 255.255.255.0 ASA1(config-network-object)# nat (inside,outside) dynamic WAN	
ASA1# show nat	ASA1# show run object
ASA1# show run nat	ASA1# show xlate
ASA1# show nat detail	ASA1# show conn
root@PC1:~# visit any website	root@PC2:~# visit any website

Dynamic NAT with Manual NAT	
ASA1(config)# clear configure nat	ASA1(config)# clear configure object
ASA1(config)# object network WAN ASA1(config-network-object)# range 192.168.114.101 192.168.114.105	
ASA1(config)# object network LAN ASA1(config-network-object)# subnet 192.168.1.0 255.255.255.0 ASA1(config)# nat (inside,outside) source dynamic LAN WAN	
ASA1# show nat	ASA1# show run object
ASA1# show run nat	ASA1# show xlate
ASA1# show nat detail	ASA1# show conn
root@PC1:~# visit any website	root@PC2:~# visit any website

Dynamic PAT with Auto NAT	
ASA1(config)# clear configure nat	ASA1(config)# clear configure object
ASA1(config)# object network LAN ASA1(config-network-object)# subnet 192.168.1.0 255.255.255.0 ASA1(config-network-object)# nat (inside,outside) dynamic interface	
ASA1# show nat	ASA1# show run object
ASA1# show run nat	ASA1# show xlate
ASA1# show nat detail	ASA1# show conn
root@PC1:~# visit any website	root@PC2:~# visit any website

Dynamic PAT with Auto NAT 2 nd Method	
ASA1(config)# clear configure nat	ASA1(config)# clear configure object
ASA1(config)# object network WAN ASA1(config-network-object)# host 192.168.114.101	
ASA1(config)# object network LAN ASA1(config-network-object)# subnet 192.168.1.0 255.255.255.0 ASA1(config-network-object)# nat (inside,outside) dynamic WAN	
ASA1# show nat	ASA1# show run object
ASA1# show run nat	ASA1# show xlate
ASA1# show nat detail	ASA1# show conn
root@PC1:~# visit any website	root@PC2:~# visit any website

Dynamic PAT with Manual NAT	
ASA1(config)# clear configure nat	ASA1(config)# clear configure object
ASA1(config)# object network LAN ASA1(config-network-object)# subnet 192.168.1.0 255.255.255.0	
ASA1(config)# nat (inside,outside) source dynamic LAN interface	
ASA1# show nat	ASA1# show run object
ASA1# show run nat	ASA1# show xlate
ASA1# show nat detail	ASA1# show conn
root@PC1:~# visit any website	root@PC2:~# visit any website

Identity NAT with Auto NAT	
ASA1(config)# clear configure nat	ASA1(config)# clear configure object
ASA1(config)# object network NO ASA1(config-network-object)# host 192.168.1.10 ASA1(config-network-object)# nat (inside,outside) static 192.168.1.10	
ASA1# show nat	ASA1# show run object
ASA1# show run nat	ASA1# show xlate
ASA1# show nat detail	ASA1# show conn
root@PC1:~# telnet 192.168.114.131	Pub-Telnet# show users

Identity NAT with Manual NAT	
ASA1(config)# clear configure nat	ASA1(config)# clear configure object
ASA1(config)# object network LAN ASA1(config-network-object)# subnet 192.168.1.0 255.255.255.0 ASA1(config-network-object)# exit	
ASA1(config)# object network WAN ASA1(config-network-object)# subnet 192.168.114.0 255.255.255.0 ASA1(config-network-object)# exit	
ASA1(config)# nat (inside,outside) source static LAN LAN destination static WAN WAN	
root@PC1:~# telnet 192.168.114.131	Pub-Telnet# show users

Policy NAT with Manual NAT	
Pub-Telnet(config)#interface loopback 1	
Pub-Telnet(config-if)#ip address 1.1.1.1 255.255.255.255	
ASA1(config)# route outside 1.1.1.1 255.255.255.255 192.168.114.131	
ASA1(config)# object network LAN	
ASA1(config-network-object)# subnet 192.168.1.0 255.255.255.0	
ASA1(config-network-object)# exit	
ASA1(config)# object network MAP	
ASA1(config-network-object)# subnet 192.168.11.0 255.255.255.0	
ASA1(config-network-object)# exit	
ASA1(config)# object network WAN	
ASA1(config-network-object)# subnet 192.168.114.0 255.255.255.0	
ASA1(config-network-object)# exit	
ASA1(config)# nat (inside,outside) source static LAN MAP destination static WAN WAN	
ASA1(config)# nat (inside,outside) source dynamic LAN interface	
ASA1# show nat	ASA1# show run object
ASA1# show run nat	ASA1# show xlate
ASA1# show nat detail	ASA1# show conn
root@PC1:~# telnet 192.168.114.131	root@PC1:~# telnet 1.1.1.1
Pub-Telnet# who	Pub-Telnet# show users

```

root@PC1:~# telnet 1.1.1.1
root@PC1:~# telnet 192.168.114.131

```

```

Pub-Telnet#who
  Line      User      Host(s)      Idle      Location
*  0 con 0           idle         00:00:00
  2 vty 0           idle         00:00:29 192.168.114.100

  Interface      User      Mode      Idle      Peer Address
Pub-Telnet#
*May 21 13:46:36.339: %SYS-5-CONFIG_I: Configured from console by console
Pub-Telnet#who
  Line      User      Host(s)      Idle      Location
*  0 con 0           idle         00:00:00
  2 vty 0           idle         00:00:06 192.168.11.10

  Interface      User      Mode      Idle      Peer Address

```